N=1, integral is an area integral of a circle using polar cordinates

$$\Omega(E) = \int_0^{2\pi}\int_0^{\infty} \delta\left(\frac{r^2}{2} - E\right) r\,dr\,d\theta$$

$$= 2\pi \int_0^{\infty} \delta\left(\frac{r^2}{2} - E\right) r\,dr$$

$$= \frac{2\pi}{\sqrt{2E}} \int_0^{\infty} \delta\left(r - \sqrt{2E}\right) r\,dr$$

$$= \frac{2\pi}{\sqrt{2E}} \sqrt{2E}$$

$$\Omega(E) = 2\pi \Rightarrow S = \ln \pi + \ln 2$$

N=2, integral is a volume integral over a 4D hypersphere.

$$\Omega(E) = \int \delta\left(\frac{r^2}{2} - E\right) d^4V$$

$$A = \frac{dV}{dr}$$ For N-dimensional sphere

$$dV = A\,dr$$

$$\Omega(E) = \int_0^{\infty} \delta\left(\frac{r^2}{2} - E\right) A_3\,dr$$

$$A_3 = 2\pi^2 R^3$$

$$\Omega(E) = 2\pi^2 \int_0^{\infty} \delta\left(\frac{r^2}{2} - E\right) r^3\,dr$$

$$\Rightarrow \Omega(E) = \frac{2\pi^2}{\sqrt{2E}} \int_0^\infty \delta(r - \sqrt{2E})\, r^3\, dr$$

$$= \frac{2\pi^2}{\sqrt{2E}} \sqrt{2E}^{\,3}$$

$$= 2\pi^2 (2E)$$

$$\Omega(E) = 4\pi^2 E$$

$$\Rightarrow S = \ln E + 2\ln\pi + \ln 4$$

$N=3$

$A_5 = \pi^3 R^5$

$$\Omega(E) = \pi^3 \int_0^\infty \delta\left(\frac{r^2}{2} - E\right) r^5\, dr$$

$$= \frac{\pi^3}{\sqrt{2E}} \int_0^\infty \delta(r - \sqrt{2E})\, r^5\, dr$$

$$= \frac{\pi^3}{\sqrt{2E}} (\sqrt{2E})^5$$

$$= \pi^3 (2E)^2$$

$$= 4\pi^3 E^2$$

$$\Rightarrow S = 2\ln E + 3\ln\pi + \ln 4$$